

Justin Bacic

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EDUCATION

Bachelor of Computer Science, Honours, Algorithms Stream Minor in Mathematics

GPA 3.74

Carleton University
Sept 2021 – Apr 2025 | Ottawa

SKILLS

Languages

Java | Python | C++ | C | JavaScript
Haskell | SQL | HTML | C# | CSS

Tools & Frameworks

Linux | Shell Scripting | Git | Flask
PostgreSQL | Qt | Node JS | Power
Automate | SharePoint | Docker

Knowledge

Data Structures | Algorithms |
Database Management Systems |
Web Development Network
Protocols | Operating Systems
Distributed Algorithms | Quantum
Computing | REST APIs | Embedded
Development | RTOS

WORK EXPERIENCE

Teaching Assistant – Programming Paradigms & Algorithms

Carleton University Sept 2024 – Apr 2025 | Ottawa

- Led office hours to **support students** struggling with concepts such as recursion improving overall academic performance
- **Developed supplementary learning material** to address areas of struggle and deepen students understanding of complex course content

Software Developer

Data Annotation Apr 2024 – Aug 2024 | Remote

- **Developed software** for training LLMs in generating **correct and readable code** to solve users' problems across **multiple programming languages**
- Debugged AI generated code for **web development, GUI design, data analysis, and mathematical reasoning** to improve model responses for diverse query topics

IT Developer Student

Transport Canada Executive Office May 2023 – Apr 2024 | Ottawa

- Led development of a **user-friendly agenda planner** with automation features that streamlined collaboration and communication among committees
- **Provided training and support to end-users**, achieving seamless adoption of the agenda planner system and positive feedback from committee leads
- Automated the staffing action approval process for the department using Power Automate, **reducing approval time by over 75%**
- **Conducted client requirement-gathering sessions** to ensure final products included all the desired functionality

PROJECTS

Machine Learning PID Controller – C, ESP-IDF, FreeRTOS, Python

Worked in a team to **design and implement a real-time system** for PID controller tuning using a genetic algorithm. **Developed embedded software** on an ESP32 microcontroller with FreeRTOS to control a wheel motor, process encoder input, and dynamically adjust motor behavior through a PID feedback loop. Developed and deployed a **genetic algorithm** that aimed to optimize PID gain values based on real-world test performance.

Delaunay Triangulation Visualization Tool – Python, Flask, Javascript, HTML/CSS

Implemented the incremental Delaunay Triangulation Algorithm in Python and **developed an interactive visualization tool** to illustrate the triangles, circumcircles and edge flips of the algorithm. Extended the project into a **Flask based web application with a JavaScript frontend** for browser-based use. **Conducted runtime experiments** across multiple input distributions to demonstrate the efficiency of randomization.

Neureset EEG Treatment Simulator – Qt/C++, Git

Collaborated in an Agile team to create a simulation of a medical device which uses brain signal input to diagnose and treat conditions. Led **sprint planning meetings**, wrote **readable and maintainable C++ code**, and **performed code reviews** for seamless integration and consistent quality across the team.

Scheduling Program – CUSERT (Carleton University Student Emergency Response Team) | Java

Created a fully automated scheduling tool using **object-oriented programming in Java** to track and coordinate each volunteer's schedule. Developed a **customized algorithm** to meet minimum staffing requirements and **reduced the time required for scheduling** from several hours to minutes.